

 **CS 391L Machine Learning**

BUILDING A RETRIEVAL-AUGMENTED GENERATION (RAG) SYSTEM FOR COURSE MATERIAL QUESTION ANSWERING

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PROBLEM & MOTIVATION

01

CONTENT OVERLOAD

Increasing volume and complexity of course materials (lecture notes, slides, textbooks)

02

LLM LIMITATIONS

Traditional LLMs fail to dynamically access domain-specific knowledge

03

CONTEXTUAL ACCURACY

Need for accurate, contextually informed responses in educational scenarios

BACKGROUND

WHAT IS RAG?

- **Retrieval-Augmented Generation:**
combines document retrieval with LLMs
- obtain relevant context from a database
- generate more accurate responses

RELATED WORK

LEWIS, 2017

*Retrieval-Augmented Generation for
Knowledge-Intensive NLP Tasks*

outperformed previous models on QA benchmarks

DONG, 2023

Adaptive AI Tutor for Any Course

significant improvements in quantitative and
qualitative assessments

GOALS

CORE GOAL

Develop a minimal viable prototype using small-scale course materials integrated with GPT-4o

01

EVALUATION

Quantitative metrics and manual review

02

STRETCH GOAL

Conduct ablation studies with different LLMs and material types

03

METHODOLOGY OVERVIEW

01 DATA PREPROCESSING

02 TEXT CHUNKING

03 EMBEDDING GENERATION
(SENTENCE-TRANSFORMERS)

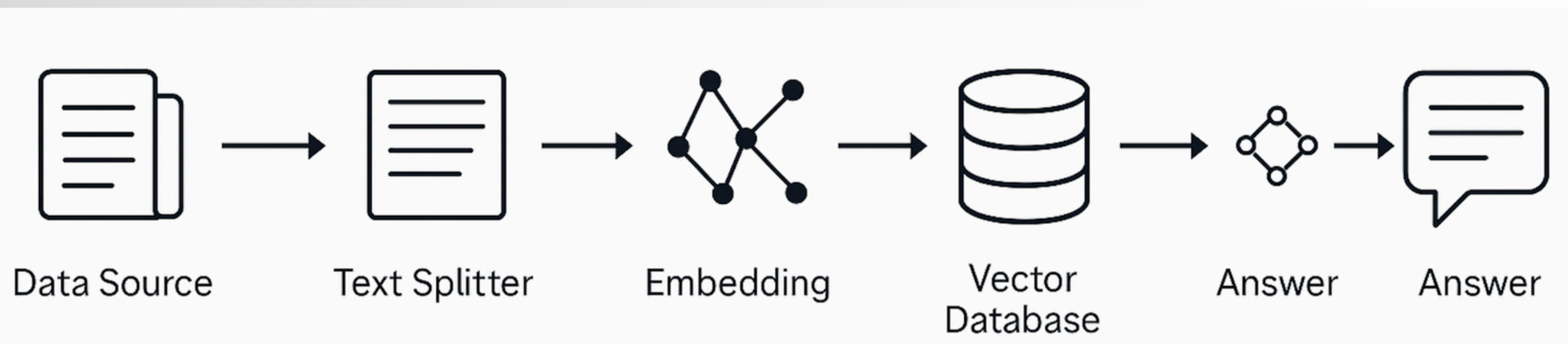
04 INDEXING (USING FAISS)

05 RETRIEVAL PIPELINE

06 PROMPT CONSTRUCTION

07 ANSWER GENERATION

08 ANSWER EVALUATION
(BLEU, ROUGEL, BERTF1)



PREPROCESSING

01

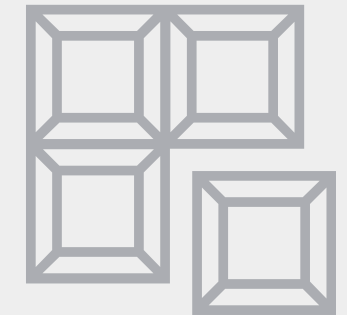
SOURCES

Lecture notes, video transcripts, textbook excerpts

02

CLEANING

Remove filler words, correct spelling errors

**03**

CHUNKING STRATEGY

Number of sentences vs. token-based chunking

EMBEDDING & INDEXING

EMBEDDING

Sentence
Transformers for
semantic
representation

01

INDEXING

FAISS for
efficient
similarity
search

02

CHOICE RATIONALE

Efficient
retrieval and
scalability

03

RETRIEVAL PIPELINE

COSINE SIMILARITY

Compared query vector to precomputed chunk embeddings

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

KEYWORD OVERLAP

Tokenized both the query and retrieved chunks. Kept top 50% of initial retrievals based on overlap

RATIONALE

Cosine similarity captures semantic similarity--focuses on the direction of vectors rather than their magnitude (L2)

Keyword overlap ensures prompt relevance

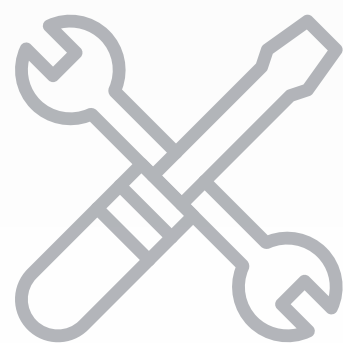
PROMPT FORMAT

You are a helpful teaching assistant answering questions. Please read the **Context** and **Question** below and try to compile a cohesive answer based on the context. Do not use your own knowledge and sticks to the context being given to you.

Context: {context_chunks}

Question: {query}

Answer:



PROMPT CONSTRUCTION

Bridge the gap between the retrieved context and the language model

Concatenate user query with top-k retrieved text chunks

Format the input to clearly separate the question and context

Experimented with a few different formats

ANSWER GENERATION

UTILIZE GPT-4O API CALL

**GENERATE INFORMED RESPONSES
BASED ON CONSTRUCTED PROMPTS**

SAMPLE Q&A

Why does the SVD of a matrix reveal its rank?

The Singular Value Decomposition (SVD) of a matrix reveals its rank because it decomposes the matrix into a series of singular values and corresponding singular vectors. The rank of a matrix is the maximum number of linearly independent columns (or rows) it has. In the SVD decomposition, each singular value corresponds to a specific singular vector (set of linearly independent vectors). The number of non-zero singular values in the decomposition corresponds to the number of these linearly independent vectors, hence revealing the rank of the matrix. Moreover, these singular values are presented in decreasing order, meaning the significant values (non-zero) come first and represent the rank of the original matrix. If a singular value is zero, it means there is a redundancy in the matrix's columns or rows.

EVALUATION METRICS

BLEU

Evaluates precision and accuracy of text

ROUGE-L

Measures overlap of content between generated and reference answers

BERTSCORE

Captures semantic similarity

Q_id	bleu	avg_rougeL	avg_bertF1
1	2.17474	0.227273	0.212875
2	2.62796	0.307692	0.204594
3	0.624266	0.0666667	0.156564
4	1.41747	0.133333	0.126541
average	2.25148	0.120441	0.0969028

CURRENT PROGRESS

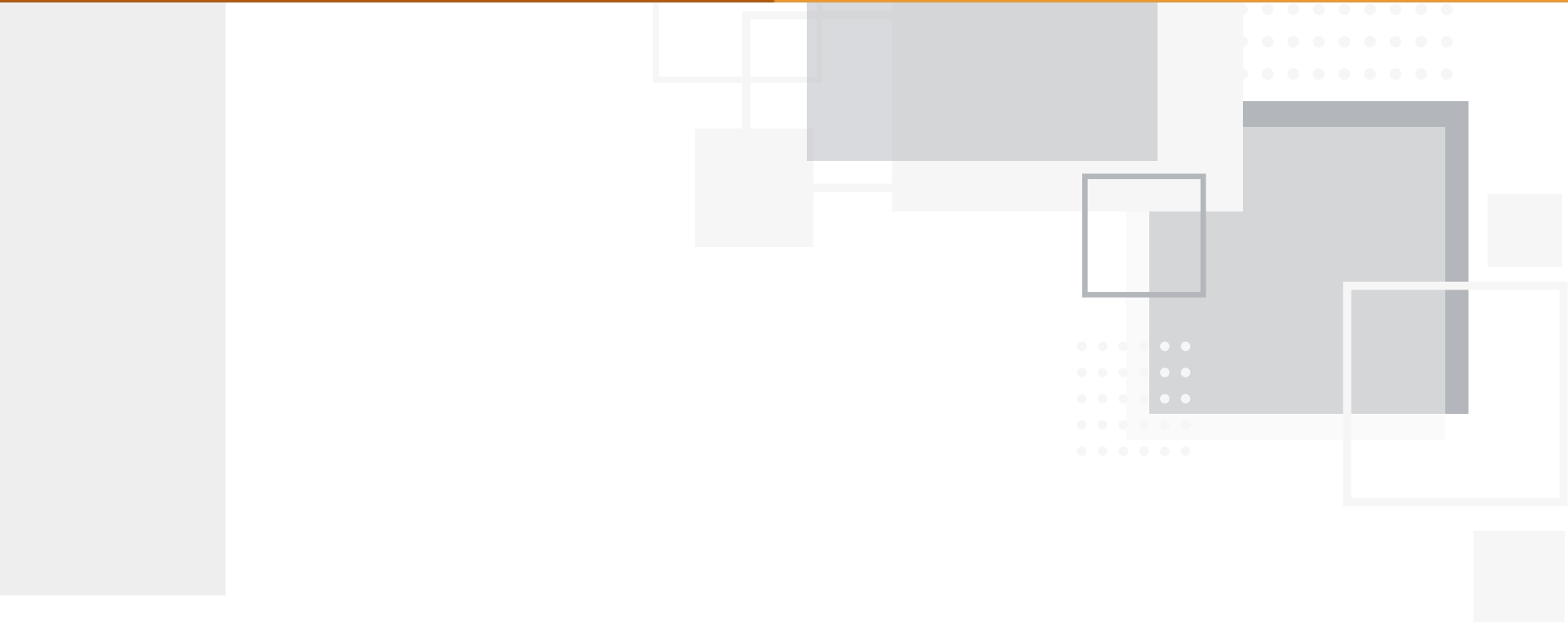
- Implemented data preprocessing and chunking
- Successfully embedded course materials
- Built FAISS index for retrieval
- Prototype of retrieval and answer generation pipeline operational
- Preliminary manual evaluation underway

CONTINUE EVALUATION

- Conduct systematic evaluation and analysis of results
- Investigate how to improve scores

PERFORM ABLATION STUDIES

- Compare system performance using GPT-3.5 vs GPT-4o
- Assess the impact of including textbook content



NEXT STEPS





THANK YOU
QUESTIONS?